# Foundations NRSG 231 · Midterm Study Guide (v4.1)

**Midterm:** Wed Oct 21, 2026, 8:30 AM, Room B137. Examplify, 90 min, 50 points. Question types: multiple choice, short answer, select all that apply, hot spot (click on a photo) and drag and drop. Arrive 15 min early and bring your phone and name badge.

**Built only from her materials:** Ch. 25 Asepsis slides, the Ch. 34 Activity lecture, Ch. 26 Vital Signs slides, Ch. 8 Communication slides, the Communication Scenario role-play deck, Weekly Skills FALL 26, her **Vital Sign Cheat Sheet**, the three **Vital Signs Extra Practice** handouts, the **BP Practice** PPT, and the Midterm FAQ "Key Content Areas" list. There is **no textbook content**. Organized by the FAQ topics, in FAQ order.

**NOT IN SLIDES** marks an FAQ item her slides don't answer. Get it from class notes or the lab checklist; this guide doesn't fill it in. The FAQ itself says **"MEMORIZE the vital sign cheat sheet."**

# ★ MUST MEMORIZE: NRSG 231 Vital Signs Ranges Cheat Sheet (her handout, verbatim)

## Blood Pressure (BP)

- Normal: Less than 120/80
- Low BP (Hypotension): Below 90 (systolic) or below 60 (diastolic)
- Elevated (Prehypertension): 120–129 (systolic) and less than 80 (diastolic)
- High BP Stage 1: 130–139 / 80–89
- High BP Stage 2: 140+ / 90+
- Hypertensive Crisis: Over 180 / Over 120  Emergency

## Temperature

- Normal: 96.4°F – 100.4°F (Think: 37°C = heaven)
- Fever (Pyrexia): Over 100.4°F (38°C = not so great)
- Hypothermia: Below 95°F (35°C = barely alive)
- Hyperthermia: Over 104°F (40°C)
- Hyperpyrexia (Super Fever): Over 106°F (41°C)  Medical Emergency

## Heart Rate (HR) – Beats per Minute (BPM)

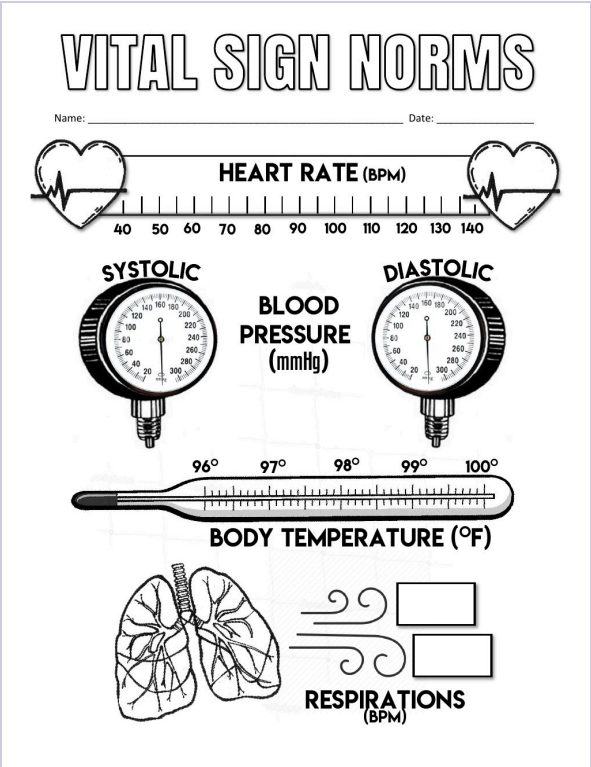
- Normal: 60–100 BPM
- Bradycardia (Too Slow): Less than 60
- Tachycardia (Too Fast): More than 100

## Respiratory Rate (Breaths per Minute)

- Normal: 12–20 breaths/min
- Bradypnea (Too Slow): Less than 12
- Tachypnea (Too Fast): More than 20

## Oxygen Saturation ($SpO_2$)

- Normal: 95%–100%
- Borderline Low: 92%–94%
- Hypoxia (Too Low): Below 92%




Vital Signs Extra Practice 2 handout: "Vital Sign Norms" (HR, BP, temp, RR)

**Cheat sheet vs her slides** (both are hers; the FAQ says to memorize the cheat sheet):
• Elevated BP: cheat sheet says diastolic **less than 80**; Slide 20 says diastolic "of 80." The cheat sheet settles the overlap.
• Stage 2: cheat sheet **140+/90+** (includes 140/90); Slide 20 says ">140, >90."
• Crisis: cheat sheet **over 180 / over 120**; Slide 20 gives only ">180."
• Normal temp: cheat sheet **96.4–100.4 °F**; Slide 5 says cells work best at 36–37.5 °C (98.6 °F).
• Hyperthermia: cheat sheet **over 104 °F (40 °C)**; Slide 7 calls >40 °C (104 °F) "dangerous." Hyperpyrexia >106 °F (41 °C) matches.

# 1 · Ch. 25 Asepsis & Infection Control

## Chain of infection and how to break it [Slides 3, 4, 14, 16]

Infection = a disease state from pathogens in or on the body. All **six links** must be present [Slide 3]:

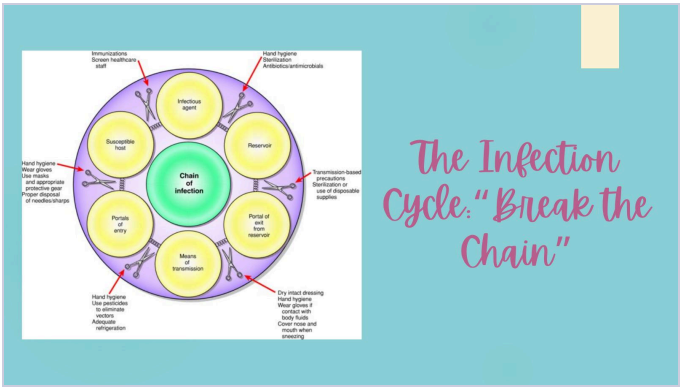
| Link | Her examples | Break the chain [Slide 4 figure] |
|---|---|---|
| 1. Infectious agent | Bacteria (E. coli, staph, strep); viruses (HIV, cold, flu, COVID-19, hepatitis A/B/C); fungi (athlete's foot, ringworm, yeast); parasites (malaria via mosquito) [Slide 5] | Hand hygiene, sterilization, antibiotics/antimicrobials |
| 2. Reservoir | People, animals, soil, food, water, milk, objects [Slide 6] | Transmission-based precautions; sterilization or disposable supplies |
| 3. Portal of exit | Respiratory, GI, GU tracts, skin breakdown, blood & drainage [Slide 11] | Dry intact dressing, hand hygiene, gloves if contact with body fluids, cover nose and mouth when sneezing |
| 4. Means (mode) of transmission | **Contact**: direct (touching, kissing, sex) or indirect via a **vector** (living creature) or **fomite** (inanimate object) [Slide 8]. **Droplet**: organisms >5 µm, heavy, land on surfaces. **Airborne**: organisms <5 µm, stay in the air [Slide 9] | Hand hygiene, pesticides to eliminate vectors, adequate refrigeration |
| 5. Portal of entry | Respiratory, GI, GU tracts, skin breakdown, blood & tissues [Slide 7] | Hand hygiene, gloves, masks and protective gear, proper sharps disposal |
| 6. Susceptible host | Multiple wounds/poor skin integrity (= multiple portals of entry), pH of GI/GU/skin, WBC count, age/sex/heredity, immunity (natural vs acquired vs passive), fatigue, nutrition, pre-existing illness, certain meds, stress, invasive or indwelling devices [Slide 12] | Immunizations; screen health care staff |

**Slide 14 (APIC poster), 5 steps:** 1 Stop at the source · 2 Block the portal of exit · 3 Interrupt mode of transmission · 4 Protect portal of entry · 5 Increase host's defenses.

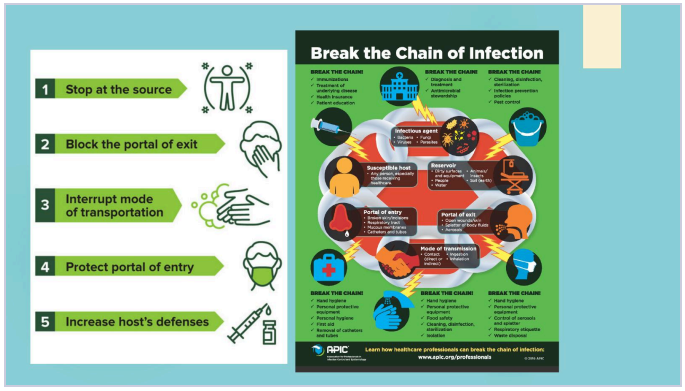
**Her practice question [Slide 16]:** breaking the chain at the *mode of transmission* = **wearing gloves during wound care**. Antibiotics act on the agent, and nutrition strengthens the host.

**Case study, Jimin [Slide 13]:** leg laceration, IV in his arm, unvaccinated for COVID, dehydration, stress and fatigue all make him a susceptible host.

**What makes an organism thrive [Slide 15]:** quantity, virulence, the patient's immune system, how long the contact lasted, and what type of contact it was.



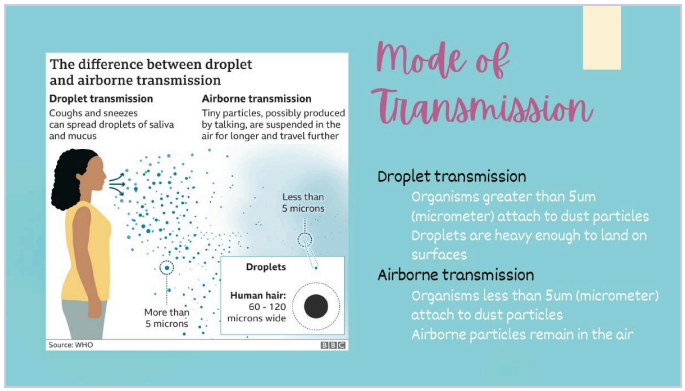

Slide 4: break the chain




Slide 14: APIC poster




Slide 8: modes of transmission (how germs spread)

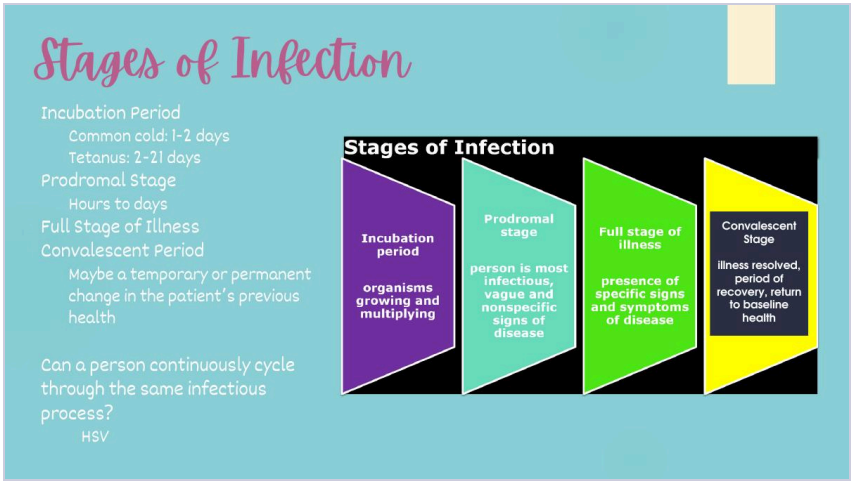



Slide 9: droplet vs airborne transmission

## Stages of infection [Slides 17–18]

1. **Incubation**: e.g. common cold 1–2 days, tetanus 2–21 days
2. **Prodromal**: hours to days; vague symptoms
3. **Full stage of illness**
4. **Convalescent**: illness resolves, recovery, return to baseline. There may be a temporary or permanent change from previous health.

Can someone cycle through the same infection again? Yes, e.g. **HSV** [Slide 17]. **Her knowledge check [Slide 18]:** a student who is "too tired to keep my head up" with a low-grade fever is in the **prodromal** stage.



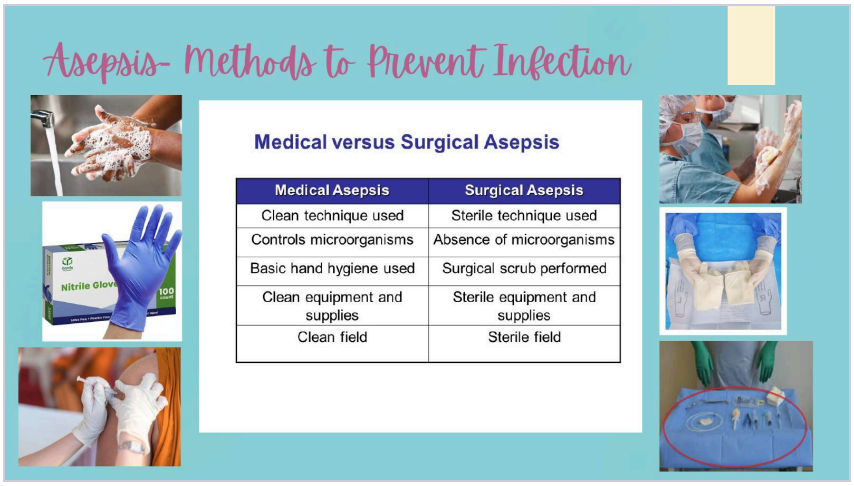

Slide 17: stages of infection

## Medical asepsis vs surgical asepsis, and asepsis practices [Slides 19–21]

| Medical asepsis: "clean technique" | Surgical asepsis: "sterile technique" |
|---|---|
| **Reduces** the number of pathogens; controls microorganisms | **Eliminates** pathogens; absence of microorganisms |
| Basic hand hygiene | Surgical scrub |
| Clean equipment and supplies; clean field | Sterile equipment and supplies; sterile field |
| Practices: hand hygiene, gloves, never put soiled linen on the floor, avoid raising dust, clean-to-dirty, put dirty linen directly in the hamper, pour contaminated liquids into the drain or "hopper" to avoid splashing | Creates and keeps a pathogen-free environment. Required for Foley catheters and central line placement/dressing changes. Sterile gloves, mask, hair net, gown, etc. |

**WHO Five Moments for Hand Hygiene [Slide 21]:** 1 Before touching a patient · 2 Before a clean or aseptic procedure · 3 After body fluid exposure risk · 4 After touching a patient · 5 After touching patient surroundings.

**NOT IN SLIDES** Slide 24 only *asks* how long to wash with soap and water, how to use sanitizer, which is more effective and when sanitizer isn't appropriate. The answers aren't on the slides.




Slide 20: medical vs surgical asepsis




Slide 21: WHO Five Moments for Hand Hygiene



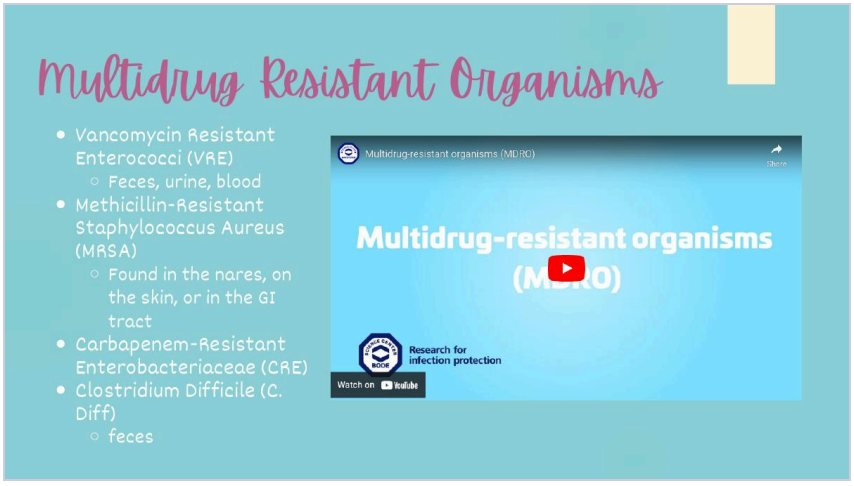

Slide 33: hand hygiene

## MDROs: what they are and how to prevent them [Slides 23, 29]

- **VRE** (vancomycin-resistant Enterococci): feces, urine, blood
- **MRSA** (methicillin-resistant Staph aureus): nares, skin, GI tract
- **CRE** (carbapenem-resistant Enterobacteriaceae)

- **C. diff** (Clostridium difficile): feces

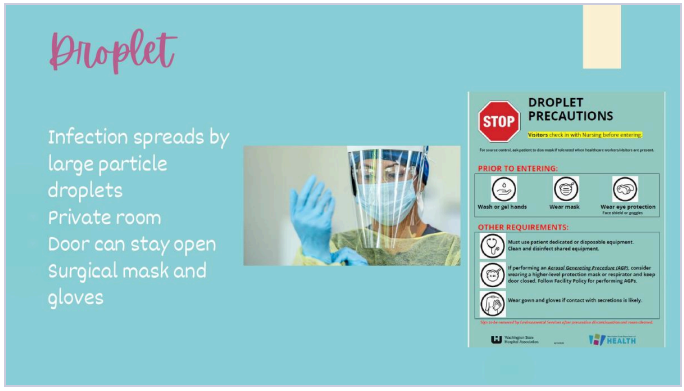
Prevention on her slides: a patient infected or colonized with an MDRO goes on **contact precautions**, with a private room, gown and gloves, glove changes after touching infective material, and equipment cleaned and not shared [Slide 29].




Slide 23: multidrug-resistant organisms (MDROs)

## Isolation precautions [Slides 26–31]

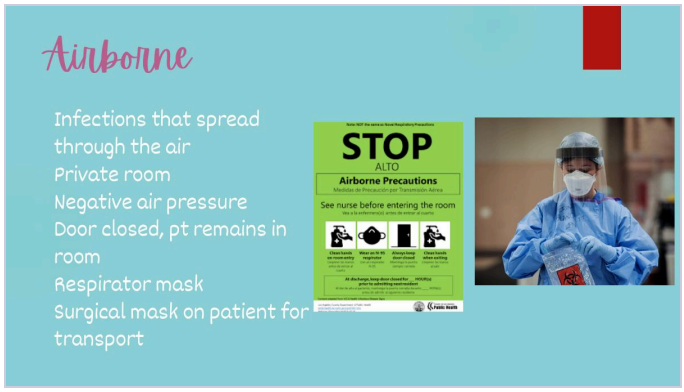
| Type | Her slide details |
|---|---|
| **Standard** [26] | All hospitalized patients regardless of diagnosis. Applies to blood, all body fluids, secretions and excretions **except sweat**, non-intact skin and mucous membranes. Newer additions: respiratory hygiene/cough etiquette, safe injection practices, and a mask for high-risk prolonged spinal canal procedures. |
| **Transmission-based** [27] | Used **in addition to** standard precautions for suspected airborne, droplet or contact pathogens. Don PPE when entering, doff when leaving. A disease can have more than one route. |
| **Droplet** [28] | Large-particle droplets · private room · **door can stay open** · surgical mask and gloves |
| **Contact** [29] | MDRO · private room · **gown and gloves** · change gloves after infective material · clean equipment, don't share |
| **Airborne** [30] | Spread through the air · private room · **negative air pressure** · **door closed**, patient stays in the room · **respirator mask** · surgical mask on the patient for transport |
| **Reverse/neutropenic** [31] | Severely immunocompromised/immunosuppressed; any cancer treatment or transplant; **WBC below 500 cells/mm³** |




Slide 28: droplet precautions




Slide 29: contact precautions




Slide 30: airborne precautions




Slide 31: reverse/neutropenic precautions

## PPE [Slides 25, 27; Weekly Skills Week 2]

PPE = **gloves, gown, masks, protective eyewear** [Slide 25]. Don PPE entering a contact or droplet room and doff when leaving [Slide 27]. Week 2 lab and ATI checklists: Performing Hand Hygiene, **Applying PPE, Removing PPE**; TracPrac: standard precautions with hand hygiene, **Don and Doff PPE** [Skills p. 1].

**NOT IN SLIDES** The donning and doffing **order** isn't on her slides or the skills sheet. Use the ATI "Applying PPE / Removing PPE" checklist from Week 2 lab.




Slide 25: PPE = gloves, gown, masks, protective eyewear

### DIAGRAM · PPE order (standard CDC sequence, not on her slides)

**DON (put on) when entering**

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| Hand hygiene | Gown | Mask / respirator | Goggles / face shield | Gloves (over gown cuffs) |

**DOFF (take off) at the door when leaving**

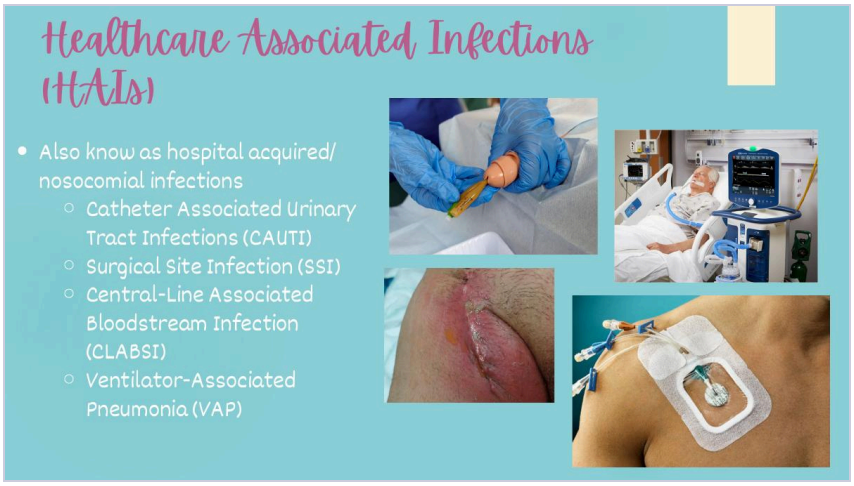
| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| Gloves (most dirty) | Goggles / face shield | Gown | Mask / respirator | Hand hygiene |

Confirm this order against your ATI "Applying PPE / Removing PPE" checklist from Week 2 lab.

Diagram (drawn, not from her materials): standard CDC donning/doffing order. Her slides and skills sheet don't give the order; check it against the ATI checklist.

## HAIs: what they are and how to prevent them [Slide 22]

Health care–associated infections, also called hospital-acquired or nosocomial infections: **CAUTI** (catheter-associated UTI), **SSI** (surgical site infection), **CLABSI** (central line–associated bloodstream infection), **VAP** (ventilator-associated pneumonia).



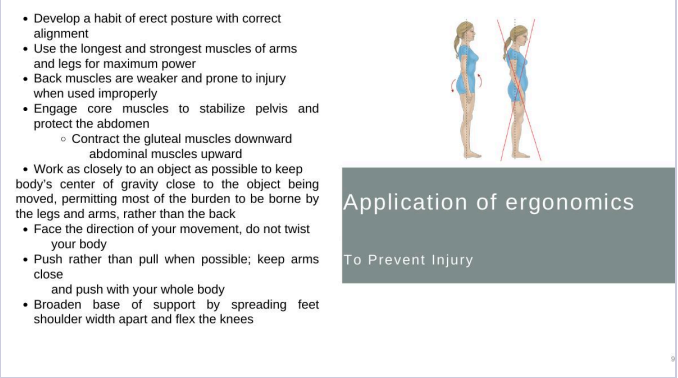

Slide 22: healthcare-associated infections (HAIs)

Tie-ins from her slides: indwelling devices make a patient more susceptible [Slide 12]. Foley catheters and central lines require **surgical asepsis** [Slide 19]. Slide 24 asks what you can conclude when a patient with a device shows infection signs. **NOT IN SLIDES** She doesn't give an HAI-specific prevention list beyond those points. Her answer to "best way to prevent infection?" (Slide 32) is shown only as a question, but the slides keep coming back to hand hygiene.

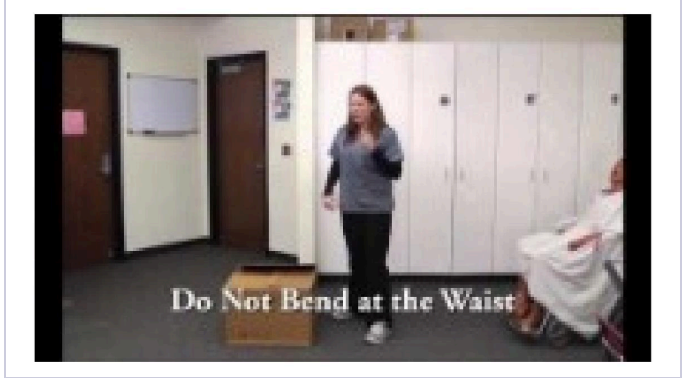
# 2 · Ch. 34 Safe Patient Handling (Activity lecture + Week 3 lab)

## Positioning & body mechanics [Slides 3–7, 12]

- **Alignment/posture**: best musculoskeletal balance with no undue strain on joints, muscles, tendons or ligaments. **Balance**: center of gravity close to a wide base of support. Spreading the feet and flexing the hips and knees increases balance and reduces strain [3].
- **Coordinated movement** uses the major, stronger muscle groups. **Postural reflexes** (inner ear, vision, autonomic contractions) maintain position and equilibrium [4].
- Factors affecting movement: developmental level; physical health (hip dysplasia, torticollis, scoliosis; dwarfism, osteoporosis, muscular dystrophy; arthritis, trauma; stroke, spinal cord injury); mental health (depression); lifestyle (diet, sedentary, smoking); culture. ADLs = eating, bathing, dressing, toileting [5].
- **Ergonomics**: designing equipment and tasks to fit the worker so injuries are prevented [6].
- **Preventing injury [7]**: erect posture; use the **longest, strongest muscles of the arms and legs** (the back is weaker); engage the core (glutes down, abs up); **work close to the object**; **face the direction of movement, don't twist**; **push rather than pull**, arms close; feet shoulder-width apart with knees flexed. Video: **do not bend at the waist** [8].

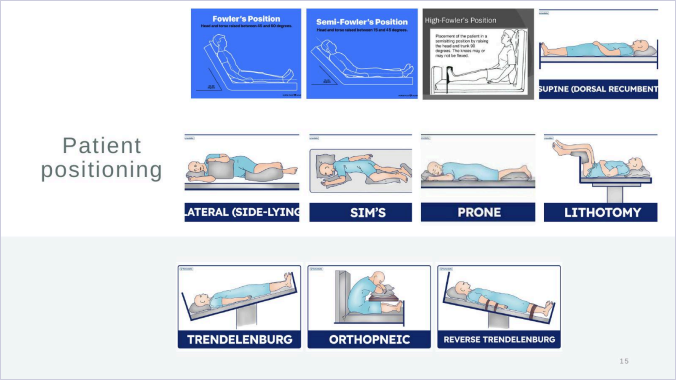



Slide 7: preventing injury (posture, body mechanics)




Slide 8 video: "Do not bend at the waist"

**Positions [Slide 12]:** Fowler's (head of bed 45–60°), semi-Fowler's (15–45°), high-Fowler's (about 90°, knees may or may not be flexed), supine (dorsal recumbent), lateral (side-lying), Sim's, prone, lithotomy, Trendelenburg, orthopneic, reverse Trendelenburg.




Slide 12: patient positioning

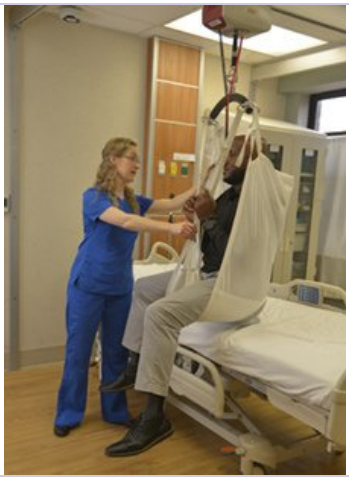



Slide 11: mobility devices

## Ensuring safe patient handling [Slides 9–11]

1. Assess the patient: ability to understand and cooperate, diagnosis and capabilities, braces/devices; encourage them to help as much as they can
2. Enough staff present
3. Clear clutter and check access and devices
4. Use a screening/assessment tool
5. Choose the equipment
6. Plan and communicate with the team and the patient
7. Give a prescribed analgesic before the move if the patient is in pain
8. Elevate the bed as needed
9. **Lock the wheels**
10. Keep the patient safe: avoid friction, support good alignment (don't grab), move in smooth rhythmic motions

**Causes of handling injuries [10]:** uncoordinated lifts, high exertion in awkward positions, manual lifting without devices, repetitive lifting and repositioning, lifting while fatigued, standing for long periods, bed-to-gurney/wheelchair transfers, repositioning in bed, and patients who are cognitively impaired, dependent, obese or unable/unwilling to help. **Devices [11]:** friction-reducing sheets, standing assist, gait belts, lateral assist device, powered full-body lifts.



Activity Slide 6: powered lift for safe patient handling



Activity Slide 6: lateral transfer with a team

## FAQ lab skills [Slide 13; Weekly Skills Week 3]

Week 3 lab: ambulation/assisted fall, wheelchairs, positioning/Hoyer lift, lateral transfers. ATI checklists: Moving and Turning Clients in Bed, Safe and Effective Transfer Techniques, Transferring Clients. TracPrac: ambulate a client; transfer to chair, commode or wheelchair; reposition in bed with safe body mechanics [Skills p. 1–2].




Activity Slide 3: pivot transfer with a gait belt and non-slip shoes

**NOT IN SLIDES** Step-by-step procedures for **ambulation with safe (assisted) fall, wheelchair transfer, lateral transfer to gurney and repositioning in bed** aren't on the slides. Slide 13 only lists them. Use the ATI checklists and lab notes.

# 3 · Vital Signs Pt. 1 & 2

The vital signs are temperature, pulse, respirations, BP, pulse oximetry and pain level [Slide 3]. **When to assess [4]:** on admission; per policy; any change in condition; loss of consciousness; before and after surgery or invasive procedures; before and after risky activity (e.g. first ambulation after surgery); before meds that affect heart or lungs; whenever the patient reports a change, pain or distress. A baseline set lets you spot changes, and a client's own normal can differ from standard values [26].

## Temperature pathophysiology [Slides 5–10]

- **Hypothalamus** controls thermoregulation by conserving heat or increasing heat loss. Temperature = heat produced minus heat lost [5].
- Affected by time of day, biological sex, age, activity, health and environment. **Core** sites: intracranial, intrathoracic, intra-abdominal. Cells work best at **36–37.5 °C (98.6 °F)** [5].
- **Circadian**: lower in the morning, **peaks 1500–1800**. Extreme cold leads to hypothermia and extreme heat to hyperthermia [6]. Slide 6 asks whether older adults run higher or lower and about women; **NOT IN SLIDES** the answers aren't given.
- **Fever (pyrexia)**: febrile vs afebrile. **>40 °C (>104 °F) dangerous**; **≥41 °C (106 °F) = hyperpyrexia, a medical emergency**. Signs: loss of appetite, headache, hot dry skin, thirst, muscle aches, fatigue [7]. Slide 8 asks which other vital sign rises with fever; **NOT IN HER MATERIALS** the answer isn't given.
- **Treating fever [9]**: fever is an immune defense, so treatment aims for comfort and preventing complications. Drugs: antibiotics for bacterial infection, antipyretics. Non-drug: cool bath, cooling blankets, removing extra clothing and blankets, oral fluids.
- **Hypothermia [10]**: poor coordination, slurred speech, amnesia, hallucinations, stupor; RR, HR and BP **decrease**. Treat by raising body temperature: warm blankets, warm fluids, Bair Hugger.




Slide 7: fever (pyrexia)




Slide 10: treating hypothermia

## All methods to measure temperature [Slides 11–12; Skills p. 2]

| Route | Key points | Time |
|---|---|---|
| **Oral** | Posterior sublingual pocket (near the carotid artery). No hot/cold drinks or smoking for 20 min before. Patient must be awake and alert. Not for small children (they may bite down). | 3 min |
| **Axillary** | Bulb in the center of the axilla, lower arm across the chest. Non-invasive and good for children but less accurate. **0.5 °C lower than oral.** | 5–10 min |

| | | |
|---|---|---|
| **Rectal** | Side-lying with the upper leg flexed; lubricated, inserted 1–1½ in (adult). Use when oral is unsafe or inaccurate (unconscious, disoriented, irrational). **0.5 °C higher than oral.** | 2–3 min |
| **Tympanic (ear)** | Close to the hypothalamus, so sensitive to core changes. **Adult: pull the pinna up and back.** Fast and easy. Cerumen impaction or otitis media distorts the reading. | 2–3 sec |

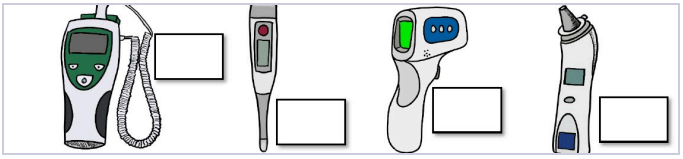
Thermometer types: infrared, tympanic, electronic oral/axillary, electronic rectal attachment [11]. The Week 4 skills sheet also lists a **temporal** video and says rectal is "discussion" only. **Extra Practice 3 handout:** temporal = taken on the **forehead**; axillary = "arm pit" with the arm in **adduction**; tympanic = **external auditory canal**; oral = **sublingual pocket using the sublingual artery**; rectal = in the rectum. The handout also asks which site is most accurate for core temp, which is affected by recent food and fluids, which shows rapid changes (near the hypothalamus), which is "non-touch," and which is least reliable. It gives no answer key, so use the slide facts above (tympanic is near the hypothalamus; no hot/cold drinks for 20 min before oral). **NOT IN HER MATERIALS** Temporal step-by-step technique.




Slide 11: thermometers




Slide 12: sites for assessing temperature



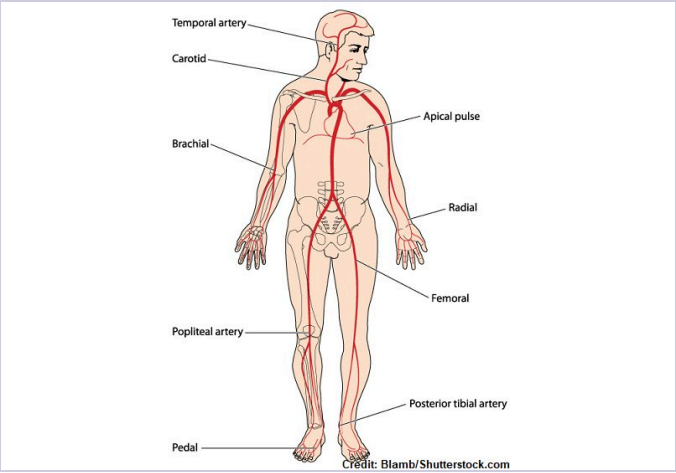
Extra Practice 3 handout: thermometer types to match to their route

## First-quarter scope for abnormal temperature

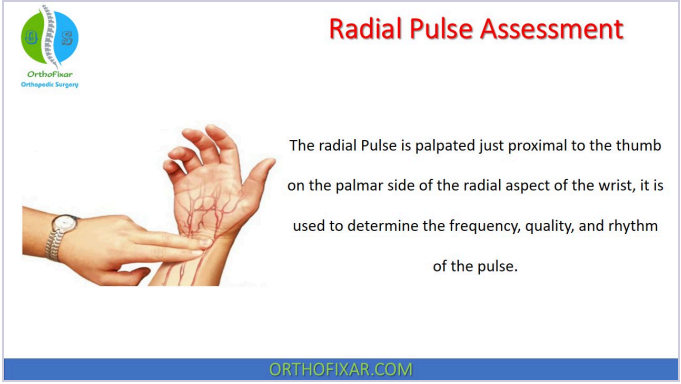
From the FAQ: most interventions are out of your scope, but you can always **recheck and notify the provider (or primary nurse)**. Her non-drug measures (cool bath, cooling blanket, removing extra blankets, oral fluids; warm blankets and fluids for hypothermia) are on Slides 9–10. **NOT IN SLIDES** She doesn't say which of these a first-quarter student may do on their own. Giving an antipyretic is medication administration.

## Full 3-part radial pulse assessment [Slides 13–15]

- The pulse is the wave of blood when the **left ventricle contracts**. Sites: carotid, brachial, radial, femoral, popliteal, posterior tibial, dorsalis pedis [13].
- Use your **2nd, 3rd or 4th fingers, never the thumb**. **Count 30 sec × 2**. If irregular, count radial for **1 full minute**, then apical for a full minute. **Apical** = PMI, auscultated for a full minute; peripheral pulses are palpated [14].
- **The 3 parts [15]: (1) Rate**, 60–100, average 80 bpm · **(2) Rhythm**, regular or irregular · **(3) Amplitude**: weak/thready, full/bounding, or imperceptible.
- **Tachycardia = more than 100; bradycardia = less than 60** [Cheat sheet]. Slide 15 leaves these blank.
- FAQ on abnormal pulse: most interventions are out of scope; **recheck the pulse and notify the provider or primary nurse**.




Slide 15 figure: pulse sites (temporal, carotid, apical, brachial, radial, femoral, popliteal, posterior tibial, pedal)




Slide 14: radial pulse assessment, finger pads on the thumb side of the wrist

## Electronic vital signs machine

Week 4 lab: the electronic VS machine for **$SpO_2$, temperature and HR**, plus manual pulse and respirations. TracPrac: pulse oximetry (intermittent or continuous), electronic BP, electronic pulse, temperature (axillary, oral, rectal-discussion), manual pulse, manual respirations, and the documentation exercise [Skills p. 2]. **NOT IN SLIDES** How to operate the machine isn't on her slides.

**Pulse oximetry [Slide 23]:** non-invasive estimate of arterial $O_2$ saturation ($SpO_2$). Probes go on the finger, ear, nose or toe. Remove thick nail polish. Wait until the reading is steady and the pulse display is at full strength. With continuous monitoring, check the skin under the probe and rotate it. **$SpO_2$: normal 95–100%, borderline low 92–94%, hypoxia below 92%** [Cheat sheet].




Slide 23: electronic vital signs machine



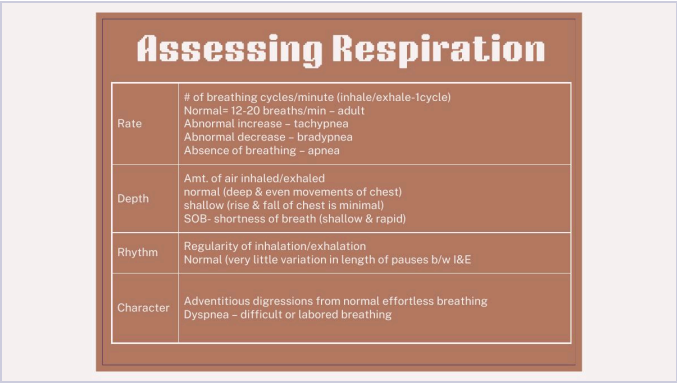

Slide 23: pulse oximetry

## Full respiratory rate assessment [Slides 16–17]

- Gas exchange of $O_2$ and $CO_2$, under both autonomic and voluntary control. Inspiration = inhalation; expiration = exhalation.
- Count **right after the pulse while still holding the wrist**, so the patient doesn't know you're counting [16].
- **Rate**: one inhale plus one exhale = 1 cycle. **Normal 12–20/min (adult)**. Tachypnea = abnormal increase, bradypnea = abnormal decrease, apnea = no breathing.
- **Depth**: normal (deep, even), shallow (minimal rise), SOB (shallow and rapid). **Rhythm**: regularity, with little variation in pauses. **Character**: **dyspnea** = difficult or labored breathing [17].
- **NOT IN SLIDES** Slide 16's blank "difficulty breathing lying down" term is never filled in on the slides.



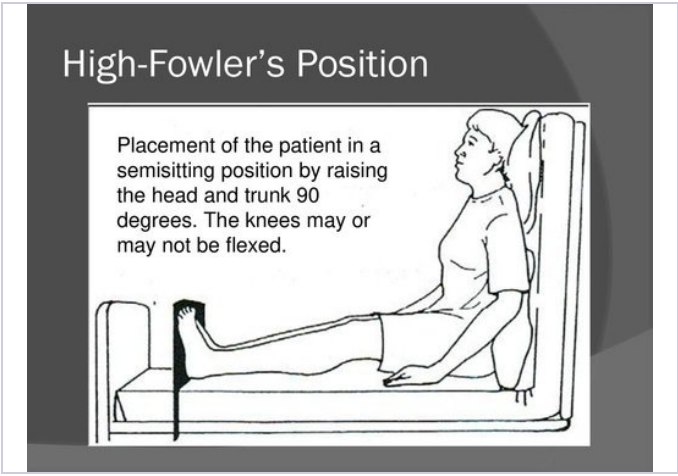

Slide 16: respirations



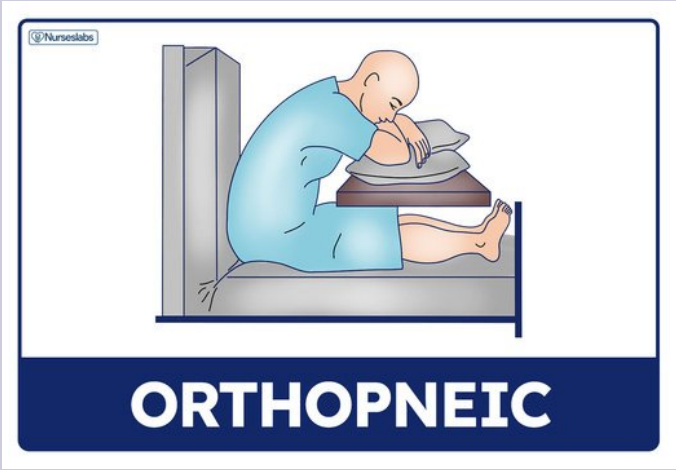

Slide 17: assessing respiration (rate, depth, rhythm, character)

## Abnormal RR: positioning, $O_2$ and scope

From the FAQ: for **SOB or dyspnea, put the patient in high-Fowler's or orthopneic position**. Both positions are pictured on Ch. 34 Slide 12; high-Fowler's = upright at about 90°. **Oxygen administration is NOT within your scope.** Reassess and notify the provider. In her Slide 25 case, the question after a high RR is which other assessment shows oxygenation, i.e. pulse oximetry [Slide 23]. **NOT IN SLIDES** The SOB-to-high-Fowler's rule and the $O_2$ scope limit come from the FAQ only, not the Vital Signs slides.




Activity Slide 12: high-Fowler's position




Activity Slide 12: orthopneic position

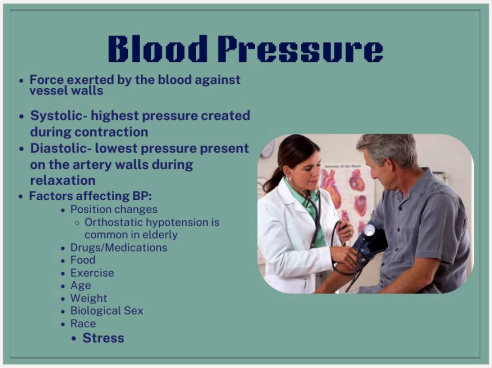
## Blood pressure [Slides 19–21]

BP is the force of blood against vessel walls. **Systolic** = highest pressure during contraction; **diastolic** = lowest pressure during relaxation. Factors: position changes (**orthostatic hypotension is common in the elderly**), drugs, food, exercise, age, weight, biological sex, race, stress [19].

| **Her BP values, exactly as on Slide 20** | |
|---|---|
| Normal | Less than 120/80 |
| Hypotension | Lower than 90 mmHg systolic or 60 mmHg diastolic |
| Prehypertension (elevated) | Systolic between 120–129 and diastolic of 80 |
| Hypertension Stage 1 | 130–139 (systolic), 80–89 (diastolic) |
| Hypertension Stage 2 | >140 (systolic), >90 (diastolic) |
| Hypertensive crisis | >180 |

> Note: these are her Slide 20 values word for word, and her exam will use them. Her wording overlaps at exactly 80 diastolic, which she lists under both "elevated" and Stage 1, and at 140/90, where Stage 2 says ">140, >90". Her cheat sheet (front page) settles them: elevated = diastolic **<80**, Stage 2 = **140+/90+**, crisis = **over 180 / over 120**.

**Technique [21]:** cuff size by **arm circumference, not age**; cuff covers **40%** of the arm circumference and the bladder covers **2/3** of the arm; arm supported **at heart level, palm up** (above the heart gives a **false low**). **Cuff too wide = false low; too narrow = false high; too loose = false high.** Korotkoff sounds = sounds of blood flowing through an artery as the cuff is deflated after occluding it. **Limb alerts:** no BP on a limb with a PICC line, an AV fistula, or on the mastectomy side.

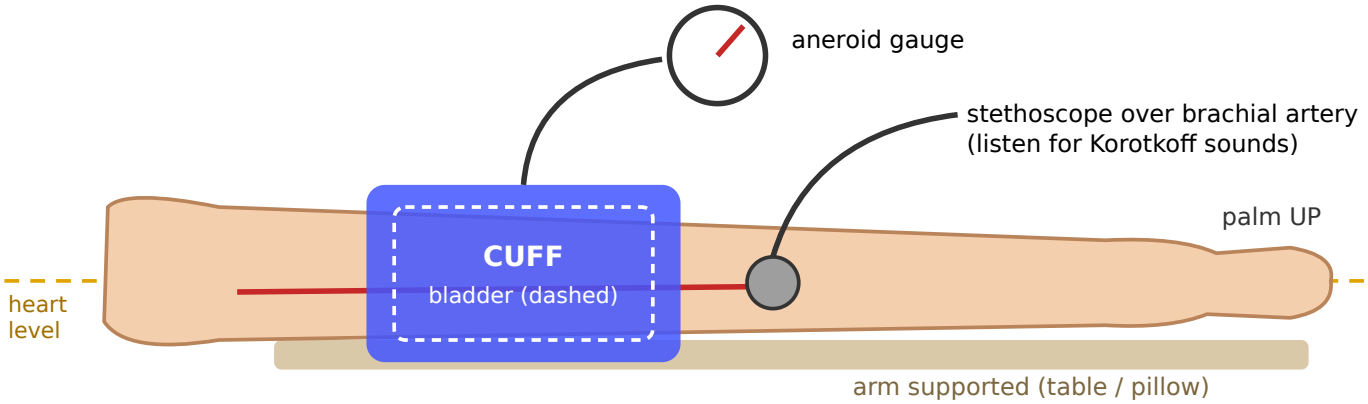



Slide 19: blood pressure



Slide 21: BP assessment with cuff and stethoscope

## DIAGRAM · Manual BP cuff placement (her Slide 21 rules)




red line = brachial artery (runs down the inner arm to the antecubital space)

- Choose the cuff by **arm circumference, not age**.
- Cuff covers **40%** of the arm circumference; the bladder covers **2/3** of the arm.
- Arm **at heart level, palm up**. Arm above the heart = **false LOW**.
- Cuff too **wide = false LOW** · too **narrow = false HIGH** · too **loose = false HIGH**.
- No BP on a limb with a PICC line, an AV fistula, or on the mastectomy side.

Diagram (drawn from her Slide 21 rules): cuff and stethoscope placement, arm at heart level

## Manual 2-step BP checklist [Weekly Skills Weeks 5–6]

Week 5 lab: **pt 1 = brachial pulse + PSP (palpated systolic pressure)**; **pt 2 = auscultation**. Week 6 = the 2-step manual BP check-off (see the remediation protocol if not passed). Practice on as many adult "patients" as you can. Manual BP goes into TracPrac only after you pass the check-off [Skills p. 3]. **NOT IN SLIDES** The full LLUSN Manual BP checklist steps aren't in these sources; use the checklist and the ATI 2-step video.

## Vital sign ranges & terminology (from her slides only)

| VS | Normal (adult) | Terms |
|---|---|---|
| Temp | 36–37.5 °C (98.6 °F) [5]; cheat sheet 96.4–100.4 °F | Fever >100.4 °F, hypothermia <95 °F, hyperthermia >104 °F [Cheat sheet]; Febrile/afebrile, pyrexia, hyperpyrexia ≥41 °C (106 °F), hypothermia, hyperthermia |

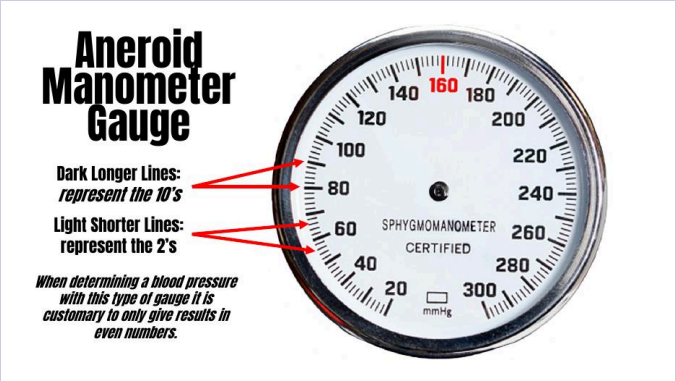
| | | |
|---|---|---|
| Pulse | 60–100, avg 80 [15] | Tachycardia >100 / bradycardia <60 [Cheat sheet], weak/thready, bounding, imperceptible, apical/PMI |
| RR | 12–20 [17] | Tachypnea >20, bradypnea <12 [Cheat sheet], apnea, dyspnea, SOB, shallow |
| BP | <120/80 [20] | Hypotension, orthostatic hypotension, elevated, Stage 1, Stage 2, crisis, systolic/diastolic, Korotkoff |
| $SpO_2$ | 95–100% [Cheat sheet] | Borderline low 92–94%, hypoxia <92% |

## Reading an aneroid gauge and BP practice [BP Practice PPT]

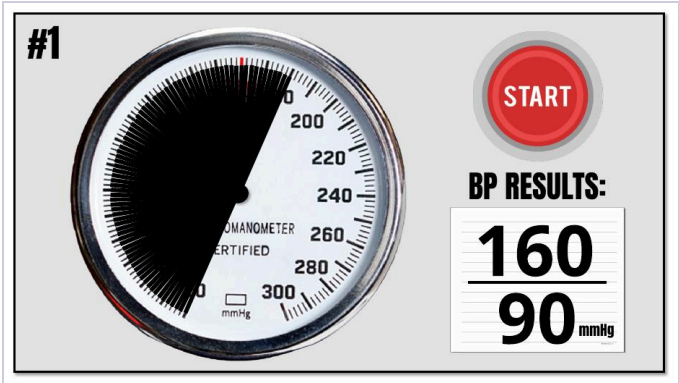
**Dark, longer lines = the 10s; light, shorter lines = the 2s.** Report BP from this gauge in **even numbers only** [BP PPT Slide 2]. Her gauge-reading practice answers: 130, 96, 52 [Slide 3]. Her BP results, classified with the cheat sheet:

| # | Reading | Cheat-sheet category |
|---|---|---|
| 1 | 160/90 | Stage 2 (140+/90+) |
| 2 | 130/60 | Stage 1 (systolic 130–139) |
| 3 | 142/84 | Stage 2 (systolic 140+) |
| 4 | 126/68 | Elevated (120–129 and <80) |
| 5 | 168/92 | Stage 2 |

The categories are my classification using her cheat sheet; the PPT shows only the readings. For mixed readings like 142/84, classifying by the higher category is my assumption; her materials don't state the rule.



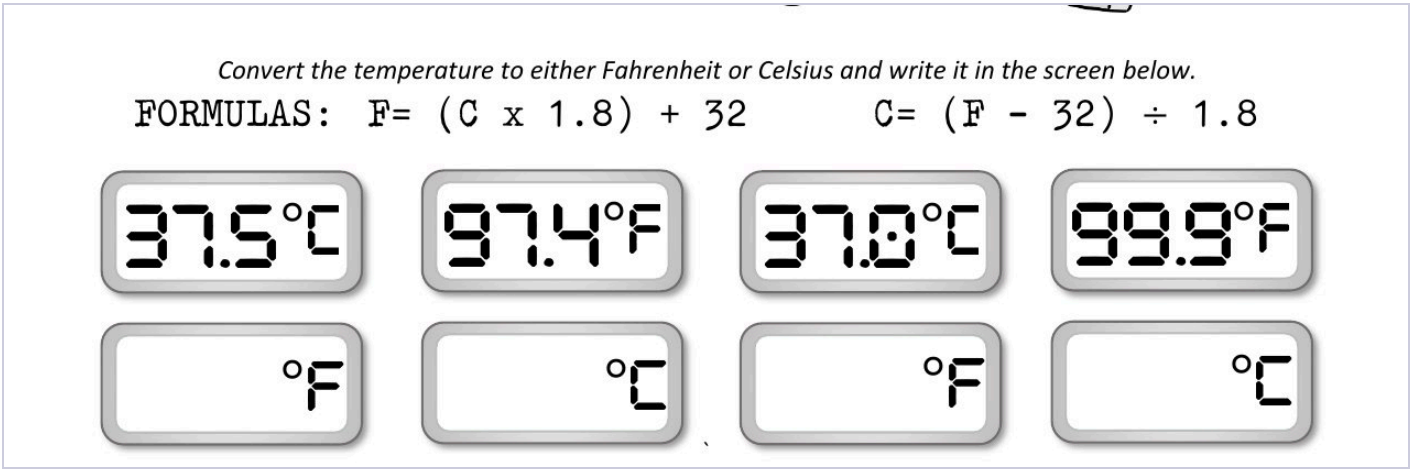

BP PPT Slide 2: aneroid gauge




BP PPT Slide 4: practice #1

## Temperature conversion [Extra Practice 3]

Her formulas: **F = (C × 1.8) + 32** and **C = (F − 32) ÷ 1.8**. Her four items, worked: 37.5 °C = **99.5 °F** · 97.4 °F = **36.3 °C** · 37.0 °C = **98.6 °F** · 99.9 °F = **37.7 °C**.



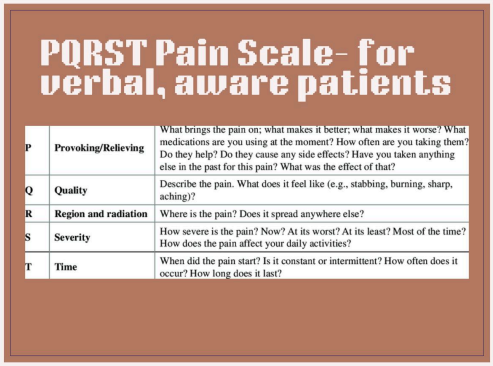

Extra Practice 3 handout: temperature conversion formulas and practice items

## PQRST pain assessment [Slide 24], for verbal, aware patients

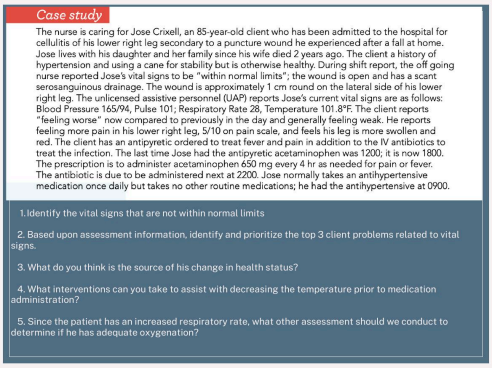
| | |
|---|---|
| **P** | Provoking/relieving: what brings it on, what makes it better or worse, current meds and their effect, past treatments |
| **Q** | Quality: stabbing, burning, sharp, aching? |
| **R** | Region and radiation: where is it, and does it spread? |
| **S** | Severity: now, at worst, at least, most of the time; effect on daily activities |
| **T** | Time: onset, constant or intermittent, how often, how long |

## ABC prioritization

The FAQ lists ABC: **airway, breathing, circulation**. **NOT IN SLIDES** No slide teaches ABC. Her Slide 25 case (Jose, 85, cellulitis: BP 165/94, P 101, RR 28, T 101.8 °F, pain 5/10) asks you to prioritize the top 3 VS problems, which is where ABC thinking applies.




Slide 24: PQRST




Slide 25: case study

# 4 · Communication (Part 1) [Ch. 8 slides + Communication Scenario deck]

## The communication process & influencing factors [Slides 3, 5–6, 13, 17]

Communication = exchanging information and generating and transmitting meaning between two or more people. It is linked to a healthy work environment (AACN) and to safe, quality care [3]. **Components [5]: sender (encoder)** starts it; **message** (visual, auditory, kinesthetic); **channel** (how it's sent); **receiver (decoder)**; **feedback** (the receiver's response back).

**Levels [13]:** intrapersonal (self-talk, which can be very damaging), interpersonal (two or more people), group (small group, organizational; group dynamics). **Influencing factors [17]:** developmental level, gender, sociocultural differences, **space and territory** (private zones: head, mouth, feet, face, neck, front of body, genitalia), roles and relationships, occupation/hobbies, physical/mental/emotional state, values, environment, **trust**.

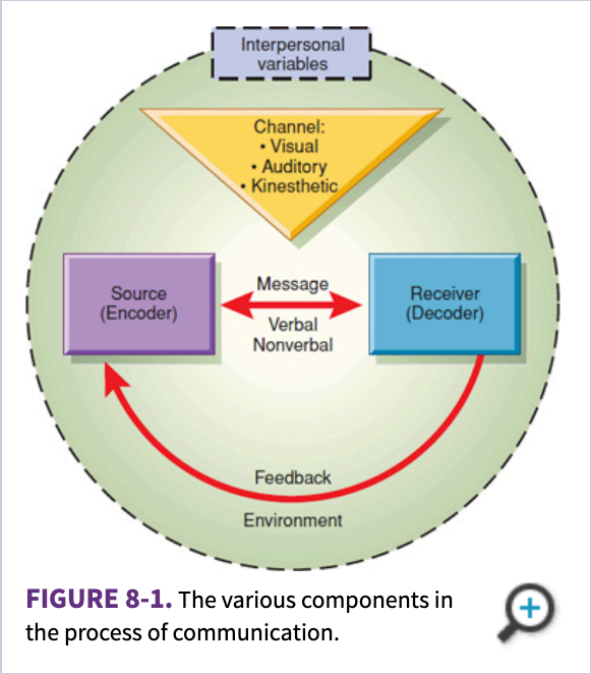



**FIGURE 8-1.** The various components in the process of communication.

Slide 6: communication process

## Nonverbal communication [Slides 7, 11–12]

Verbal = spoken and relies on language. For non-English speakers, an interpreter is the priority. **Nonverbal**: touch, eye contact, facial expression, posture, gait, gestures, appearance/grooming, sounds, silence. **If verbal and nonverbal conflict, believe the nonverbal.** Other forms: electronic (social media, email, text) and paper. Her practice question [12]: stooped gait, grimacing and gasping point to **pain**.




Ch. 8 Slide 11: other forms of communication: electronic (text message)

## Interpersonal competencies [Slides 21–22, 24]

| Social relationship [21] | Helping (professional) relationship [22] |
|---|---|
| Mutual, spontaneous, both people's needs considered, no time limit, information shared equally | Meets the patient's needs, has a specific purpose and goals, three phases, a definite end, **unequal sharing of information** |

**Dispositional traits [24]:** warmth/friendliness, openness, respect, empathy, honesty, authenticity, trustworthiness, caring, competence. **Empathy** = seeing things from the patient's point of view. **Sympathy** shifts the focus to the nurse, whose own feelings get projected onto the patient.



Ch. 8 Slide 21: social relationship



Ch. 8 Slide 22: helping (professional) relationship

## Phases of the therapeutic (helping) relationship [Slide 23]

| | |
|---|---|
| **Orientation** | The patient calls the nurse by name and describes each person's role. Agree on goals, the location, frequency and length of contacts, and how long the relationship lasts. |
| **Working** | The patient actively takes part, cooperates toward mutually acceptable goals, and expresses feelings and concerns. |
| **Termination** | The patient helps identify goals met or progress made and talks about feelings about ending. |

## Effective vs ineffective communication [Slides 25, 32–33, 40]

| Therapeutic ✓ [Slide 25] | Non-therapeutic ✗ [Slide 33] |
|---|---|
| Conversation skills (tone, clarity, truthfulness, open mind); active listening (sit, alert, relaxed, eye contact if appropriate); **silence**; touch (shoulder or hand, otherwise ask consent); humor used carefully. Questions: **open-ended** ("How are you feeling right now?"), **validating** ("You seem upset, can we talk about it?"), **clarifying** ("Did I understand you correctly…?"), **reflective** ("What went well today?"). Focus on feelings; **state observed behaviors** ("I can see you have been crying…"); reflect/restate/rephrase ("It sounds like you're frustrated…"); neutral responses; **exploring** ("Let's talk about it"). | Closed-ended questions; forgetting they are human; failure to listen; **leading questions** ("You're not really sad, right?"); **giving advice**; **changing the subject**; **talking about yourself** ("My grandma also had cancer"); telling the patient they're wrong; **false reassurance**; **clichés** ("Everything happens for a reason"); **giving approval** ("I think you are making the right choice"); **asking why**; **defending other staff** ("Sally is actually a fantastic nurse…"). |

Her practice questions: the patient is silent after hearing a diagnosis, so **allow time to think** [32]. "I am sure everything will be fine" = **false reassurance** [40].

## Role-play scenarios: technique vs barrier [Communication Scenario deck]

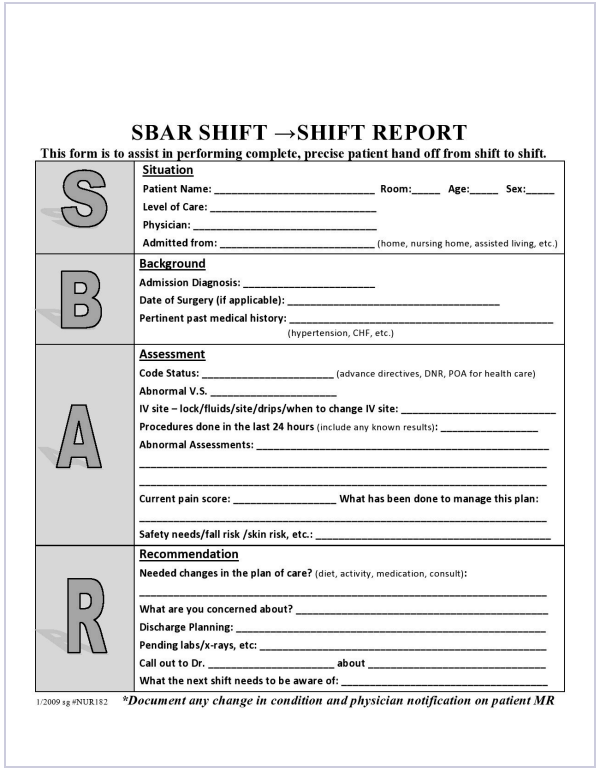
| # | Patient | Therapeutic | Non-therapeutic |
|---|---|---|---|
| 1 | Angry, long ED wait | Reflective question/comment | Aggressive behavior |
| 2 | Nervous, IV start | Gentle, appropriate humor | Arguing |
| 3 | Hopelessness | Exploring statement (assess safety) | Failure to listen |
| 4 | Doesn't understand plan | Open-ended questions | Accusatory "why" questions |
| 5 | Anxious pre-op | Realistic reassurance and support | False reassurance/promises |
| 6 | Frustrated, chronic HTN | Active listening | Judgmental comments |
| 7 | Grieving family member | Emotional/spiritual support | Clichés |
| 8 | Confused elderly | Presenting reality, reorientation | Minimizing concerns |
| 9 | Isolated adolescent | Restating/mirroring | Nonverbal boredom |
| 10 | Refusing treatment | Clarifying statement | Disapproving comments |
| 11 | Overwhelmed new parent | Validating statement | Giving advice |
| 12 | Unexpected post-op pain | Direct, focused question | Leading question |
| 13 | Teen, body image | Empathetic statement | Intrusive/premature probing |
| 14 | Seriously ill, afraid of dying | Therapeutic touch, with permission | Talking about yourself |
| 15 | Behavior doesn't match pain | Making observations | Ignoring nonverbal cues |
| 16 | Transgender, respect | Affirming identity | Dehumanizing/dismissive |
| 17 | Parent of chronically ill child | Therapeutic silence | Only yes/no questions |
| 18 | Student near burnout | Collaborative problem-solving | Gossip/rumors |
| 19 | Parent learning diabetes care | Offering self and support | Poor professional presentation |
| 20 | Limited English | Qualified medical interpreter | Frustrated/dismissive |

## Interventions for impaired communication [Slides 41–43]

- **Visually impaired**: announce yourself and give your name; speak in a normal tone; explain before touching; say when you're leaving; call light within reach; orient them to room sounds and layout; glasses clean, contacts in.

- **Hearing impaired**: get their attention first (gentle touch or move into view); face them; simple sentences at a natural pace for lip reading; no gum or covering your mouth; pantomime, sign or write; hearing aids clean, working and in place.
- **Physical barrier (laryngectomy, ET tube)**: simple means such as eye blinks or hand squeezes for yes/no, writing pads, communication boards, flash cards, sign language; make sure everyone can use the method; be patient; make sure they can signal for help (call bell).
- **Cognitively impaired**: eye contact; quiet setting; simple, concrete steps without pronouns or abstract terms; **avoid open-ended questions and offer choices** ("brown pants or gray?"); if there's no response after **2 minutes**, repeat, then take a break.
- **Unconscious**: hearing is the **last sense lost**, so assume they can hear; talk normally; speak before touching; keep noise low.
- **Non-English speaking**: **use an interpreter whenever possible**; a translation dictionary; simple sentences in a normal tone; pantomime; watch nonverbal cues.

**SBAR [Slide 44]:** Situation, Background, Assessment, Recommendation. Used for hand-off and nurse-to-provider communication to make communication more effective.



**SBAR SHIFT →SHIFT REPORT**

**This form is to assist in performing complete, precise patient hand off from shift to shift.**

**S — Situation**

Patient Name: _____ Room:_____ Age:_____ Sex:_____
Level of Care: _____
Physician: _____
Admitted from: _____ (home, nursing home, assisted living, etc.)

**B — Background**

Admission Diagnosis: _____
Date of Surgery (if applicable): _____
Pertinent past medical history: _____
(hypertension, CHF, etc.)

**A — Assessment**

Code Status: _____ (advance directives, DNR, POA for health care)
Abnormal V.S. _____
IV site – lock/fluids/site/drips/when to change IV site: _____
Procedures done in the last 24 hours (include any known results): _____
Abnormal Assessments: _____
_____
_____
Current pain score: _____ What has been done to manage this plan:
_____
Safety needs/fall risk /skin risk, etc.: _____

**R — Recommendation**

Needed changes in the plan of care? (diet, activity, medication, consult):
_____
What are you concerned about? _____
Discharge Planning: _____
Pending labs/x-rays, etc: _____
Call out to Dr. _____ about _____
What the next shift needs to be aware of: _____

1/2009 sg #NUR182 **Document any change in condition and physician notification on patient MR*

Ch. 8 Slide 45: SBAR shift report form

Her Week 4 Wrap Up announcement says the Communication PPT is midterm material even though it wasn't covered in class.

## Gaps: FAQ items her materials don't answer

- Hand hygiene times and when sanitizer isn't appropriate (Slide 24 asks without answering)
- PPE donning/doffing order
- HAI prevention specifics beyond hand hygiene, asepsis for Foleys and central lines, and device risk
- Step-by-step: ambulation with safe fall, wheelchair transfer, lateral transfer to gurney, repositioning in bed
- Temporal technique steps (the handout only says "forehead"); older-adult and women temperature questions (Slide 6); which VS rises with fever (Slide 8); answer key for Extra Practice 3 items 6–10
- Extra Practice 1 blanks with no answers in her materials: Cheyne-Stokes, rales, VS documentation order, the 10-second pulse example
- First-quarter scope for abnormal temperature (FAQ: recheck and notify)
- The "difficulty breathing lying down" term (blank on Slide 16)
- Electronic VS machine operation
- High-Fowler's/orthopneic for SOB and $O_2$ out of scope (FAQ only)
- LLUSN manual 2-step BP checklist steps
- ABC prioritization

# Gaps her materials don't cover — standard nursing answers (verify against her checklists)

> These answers are **not from her materials**. They come from standard references (CDC, WHO, AHA, NIOSH, the Open RN *Nursing Skills* open textbook and device manuals), and **none are from Taylor**. If her ATI or LLUSN checklist says something different, go with her checklist.

## 1. Hand hygiene: times and when sanitizer is not appropriate [Slide 24]

- **Alcohol rub:** a palmful covering all surfaces, rub **~20 sec until dry**. This is the preferred method for most clinical care because it kills more germs and is faster.
- **Soap & water:** wet hands, lather and scrub **at least 20 sec** (all surfaces, between fingers, thumbs, under nails), rinse, dry with a towel, and turn off the faucet with the towel.
- **Use soap & water, NOT sanitizer, when:** hands are **visibly soiled**; caring for a patient with **C. diff** or **norovirus** (alcohol doesn't kill C. diff spores); before eating; after using the restroom.
- **Best way to prevent infection [Slide 32]:** hand hygiene.

Source: CDC, Hand Hygiene in Healthcare Settings; WHO Five Moments.

## 2. PPE donning & doffing order

| Don (outside the room) | Doff (at the door / anteroom) |
|---|---|
| 1 Hand hygiene → 2 **Gown** (tie at neck and waist) → 3 **Mask or respirator** (mold the nose piece; fit-check an N95) → 4 **Goggles / face shield** → 5 **Gloves** (pulled over the gown cuffs) | 1 **Gloves** (most contaminated; glove-to-glove, then skin-to-skin) → 2 **Goggles / face shield** (handle by the band) → 3 **Gown** (pull away from the body and roll it inside-out) → 4 **Mask/respirator** (by the ties or straps; take off an N95 **after leaving** the room) → 5 **Hand hygiene** |

- Do hand hygiene right away if your hands get contaminated at any step. CDC also accepts removing the gown and gloves together.

Source: CDC, "Sequence for Putting On / Removing PPE" (CS250672).

## 3. HAI prevention specifics [Slides 22, 24]

- **All HAIs:** hand hygiene, standard/transmission precautions, clean equipment, and remove devices as soon as they aren't needed.
- **CAUTI:** insert only when indicated, using sterile technique; keep a closed system; keep the bag **below the bladder and off the floor**; no kinks; do daily perineal care; review the need daily and remove early.
- **CLABSI:** sterile insertion with chlorhexidine (CHG) skin prep; **"scrub the hub"** before every access; keep the dressing clean, dry and intact, and change it per policy; review the need daily.
- **SSI:** pre-op CHG bathing; clip hair, don't shave; keep the dressing sterile; monitor glucose; watch for redness, drainage and fever.
- **VAP: HOB 30–45°**, frequent oral care, suctioning with good technique; daily sedation and weaning checks are done by the provider.
- **Slide 24, "patient with a device shows infection signs":** suspect the device as the source (an HAI). Report it, and the device may need to be removed or cultured.

Source: CDC/HICPAC guidelines for CAUTI, intravascular catheter infections, SSI; IHI ventilator bundle.

## 4. Ch. 34 lab skills: step by step

**Before every move:** hand hygiene, two identifiers, explain the move, enough staff, bed locked at a working height, non-skid footwear, and a gait belt if the patient can bear weight.

- **Ambulation with a safe (assisted) fall:** dangle the patient on the edge of the bed and check for dizziness. Walk on the **weak side, slightly behind**, with an underhand grip on the gait belt. If the patient starts to fall: **don't try to hold them up**. Widen your stance, pull them close against your hip, and **slide them down your leg to the floor** while protecting their head. Stay with them, call for help, assess them before moving them, and report it.
- **Bed → wheelchair (stand-pivot):** put the chair on the patient's **strong side at about 45°** to the bed, **lock the wheels** and swing the footrests away. Lower the bed. The patient sits at the edge with feet flat. Face the patient, hold the gait belt, bend your knees and brace their knees. "On 3" they stand, **pivot** toward the strong side, and sit back once the chair touches the backs of their legs. Then put the footrests back, position them and leave the call light in reach.
- **Lateral transfer bed → gurney:** use **at least 2–3 staff** and a **friction-reducing sheet or slide board**. Put the bed flat, line the gurney up level with the bed (or slightly lower) and **lock both**. Logroll the patient to place the device. "On 3," **pull rather than push or reach** while keeping your back straight. Then raise the rails and secure the straps.
- **Repositioning up in bed:** lay the bed flat at **waist height**, lock it, and use 2 staff with a friction-reducing/draw sheet. The patient bends their knees, tucks their chin and crosses their arms. Use a wide stance, and "on 3" shift your weight from your back leg to your front leg. Turning/side-lying: logroll, then use pillows behind the back and between the knees. Reposition at least **every 2 h**.

Source: NIOSH/ANA Safe Patient Handling; Open RN *Nursing Skills* (2nd ed.), Ch. 3 Mobility.

## 5. Temperature questions she leaves open

- **Temporal technique:** forehead clean and dry, hair pushed back. Hold the button and **slide straight across the forehead, midline to the hairline**. If the patient is sweating, keep holding and touch behind the earlobe. Release the button and read. (Exergen instructions)
- **Older adults [Slide 6]:** run **lower** (often about 96.8 °F / 36 °C) and have a weaker fever response. A "normal" temp can still be an infection.
- **Women [Slide 6]:** temperature changes with hormones. It's about **0.3–0.6 °C higher after ovulation** (progesterone), and menopause can bring hot flashes.
- **Which VS rises with fever [Slide 8]: pulse**, about 10 bpm for every 1 °C, and **respirations** (higher metabolic rate).
- **Extra Practice 3, items 6–10:** 6 most accurate core = **R** · 7 affected by food/fluids = **O** · 8 rapid changes near the hypothalamus = **TY** · 9 "non-touch" = **TE** · 10 most unreliable = **A** (axillary).
- **First-quarter scope for abnormal temp: recheck** (same or another route), **notify** the primary nurse or provider, and document. With the nurse's OK you can do comfort measures: remove extra blankets, apply a cool cloth, offer oral fluids if there's no restriction (warm blankets for low temp). **No antipyretics**, because that's medication administration.

Source: Open RN *Nursing Skills* Ch. 1 Vital Signs; manufacturer instructions.

## 6. Respiration terms & Extra Practice 1 blanks

- **Orthopnea** = difficulty breathing **lying down**, relieved by sitting up (the blank on Slide 16).

- **Cheyne-Stokes** = cycles of breaths that get deeper then shallower, followed by **periods of apnea** (seen near death, with brain injury or heart failure).
- **Rales (crackles)** = fine popping/crackling sounds, usually on inspiration, from **fluid in small airways or alveoli**.
- **Documentation order: T, P, R, BP**, then $SpO_2$ and pain. Note the route and anything unusual.
- **Ivan example:** 13 beats × 6 = **P 78**; **R 16**; **T 100.6 °F oral**, which is a fever by her cheat sheet (>100.4). Write it as "T 100.6 °F (O), P 78, R 16."
- **SOB/dyspnea:** high-Fowler's (about 90°) or orthopneic (leaning forward over the overbed table) lets the lungs expand better. **$O_2$ is a medication and needs a provider order**, so a student reassesses and notifies.

Source: Open RN *Nursing Skills* Ch. 1 Vital Signs & Ch. 11 Respiratory Assessment.

## 7. Electronic VS machine (e.g. Welch Allyn Spot/Connex)

1. Hand hygiene; **clean the machine**; identify the patient and explain.
2. BP: choose the cuff by arm circumference and wrap it over the brachial artery (match the artery marker), with the arm at heart level. Press **Start** and have the patient keep still and not talk.
3. $SpO_2$: put the probe on a **warm finger with no nail polish** and wait for a steady waveform/pulse reading. The machine's HR is pulled from the probe.
4. Temp: put on a **probe cover**. **Blue probe = oral/axillary, red probe = rectal.** Wait for the beep.
5. **Check any abnormal reading manually**, and always check an irregular pulse manually (count apical for 1 full min). Count RR yourself, because the machine doesn't measure it.
6. Document, then clean the equipment and return it to the charger.

Source: Welch Allyn Connex Spot / Spot Vital Signs user manuals; Open RN *Nursing Skills* Ch. 1.

## 8. Manual 2-step BP (palpated systolic, then auscultated)

1. **Prep:** no caffeine, exercise or smoking for 30 min; empty bladder; seated with the **back supported, feet flat, legs uncrossed**; **rest 5 min**; bare arm supported at **heart level**; no talking.
2. **Cuff:** right size; bladder centered over the **brachial artery**; lower edge about **2–3 cm (1 in) above the antecubital fossa**.
3. **Step 1, palpated systolic (PSP):** feel the radial (or brachial) pulse and inflate until it **disappears**. That number is the estimated systolic. Deflate fully and **wait 30–60 sec**.
4. **Step 2, auscultate:** put the stethoscope over the brachial artery and inflate to **30 mmHg above the PSP**. Deflate at **2–3 mmHg per second**.
5. **First clear Korotkoff sound = systolic**; **sounds disappear = diastolic**. Keep listening 10–20 mmHg below the last sound, then deflate fully.
6. Record in **even numbers** (her gauge rule) with the arm and position. Wait **1–2 min** before repeating. Recheck and notify if abnormal.

Source: AHA Scientific Statement on BP Measurement (Muntner et al., *Hypertension* 2019); 2017 ACC/AHA BP guideline.

## 9. ABC prioritization

- **A**irway → **B**reathing → **C**irculation. A problem that threatens life now comes first. Other rules: **acute over chronic**, **actual over potential**, and physiological needs first (Maslow).

- **Her Slide 25 case (Jose):** by ABC, **RR 28** (breathing) is first; then circulation: **P 101** and **BP 165/94**. **T 101.8 °F** also needs follow-up (cellulitis, so an infection is spreading). Recheck, check $SpO_2$, and notify.

Source: standard NCLEX prioritization frameworks (NCSBN clinical judgment model; Open RN *Nursing Fundamentals* Ch. 4).